

Empirical Software Engineering

Write your answers directly on these pages; there is always a risk that loose papers disappear. Use the back also if possible.

On 7 February at 10.00–11.00 you are welcome to room 4123 in the EDIT house (Johanneberg), with questions about the grading. Bring a copy of the exam with you.

Richard will be at the written exam after 10.00 to answer any questions.

“I have no special talent. I am only passionately curious.”

— Albert Einstein

Grade 3: 33 points; ~50%

Grade 4: 46 points; ~70%

Grade 5: 59 points; ~90%

Maximum: 66 points

Question 1 :

(12p) Conducting Bayesian data analysis with a focus on causal analysis requires many steps. Some of the key steps are (i) designing a structural causal model (i.e., DAG), (ii) conducting prior predictive check, and (iii) conducting posterior predictive checks.

List and explain what activities you perform **before** you do each step above (*i-iii*). (6p)

Explain how each step above (*i-iii*) is conducted (e.g., what you do when you conduct a prior predictive check) and what the purpose is with that step (i.e., why you do it). (6p)

Question 2 :

(6p) To deal with overfitting we can apply several different strategies. One strategy is to design models that through their design are adapted to deal with this challenge by applying partial pooling.

Use math notation and provide **three** design of models where we use complete pooling, no pooling, and partial pooling.

Question 3 :

(8p) What is the **purpose** and **limitations** of using *Formal Theory* and *Field Experiments* as a research strategy? Provide **examples**, i.e., methods for each of the two categories, and **clarify** if one use mostly qualitative or quantitative approaches (or both).

Question 4 :

(4p) Below follows an abstract from a research paper. Answer the questions,

- Which of the eight research strategies presented in the ABC framework does this paper likely fit? **Justify and argue!**

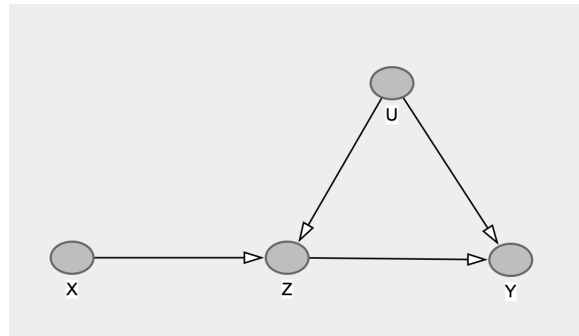
Context In software development organizations employing weak or collective ownership, different teams are allowed and expected to autonomously perform changes in various components. This creates diversity both in the knowledge of, and in the responsibility for, individual components.

Objective Our objective is to understand how and why different teams introduce technical debt in the form of code clones as they change different components.

Method We collected data about change size and clone introductions made by ten teams in eight components which was part of a large industrial software system. We then designed a Multi-Level Generalized Linear Model (MLGLM), to illustrate the teams' differing behavior. Finally, we discussed the results with three development teams, plus line managers and the architect team, evaluating whether the model inferences aligned with what they expected. Responses were recorded and thematically coded.

Results The results show that teams do behave differently in different components, and the feedback from the teams indicates that this method of illustrating team behavior can be useful as a complement to traditional summary statistics of ownership.

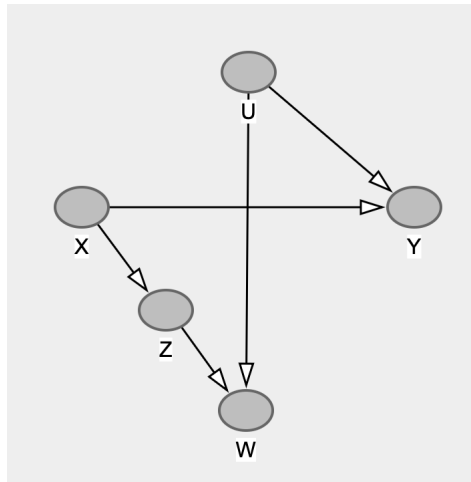
Conclusions We find that our model-based approach produces useful visualizations of team introductions of code clones as they change different components. Practitioners stated that the visualizations gave them insights that were useful, and by comparing with an average team, inter-team comparisons can be avoided. Thus, this has the potential to be a useful feedback tool for teams in software development organizations that employ weak or collective ownership.

**Question 5 :**

(7p) See the DAG above.

We want to estimate the direct **and** total causal effect of X on Y . What is the minimum adjustment set in the two cases? (2p)

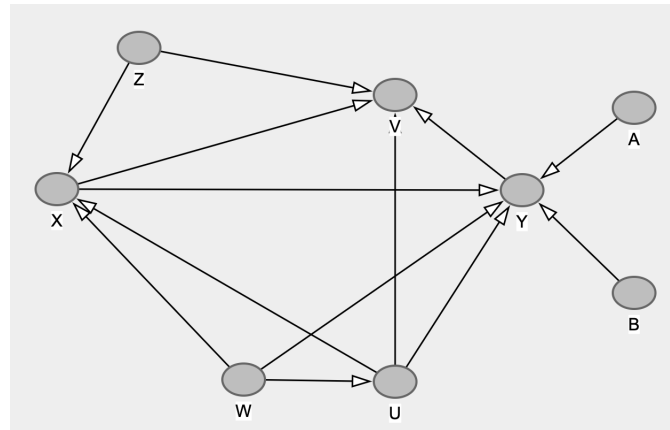
Design a **complete** model, in math notation, for the **direct** effect. The outcome Y is a count $0, \dots, \infty$. Add, what you believe to be, **suitable priors on all parameters**. State any assumptions concerning your likelihood and the variables you use! (5p)

**Question 6 :**

(7p) See the DAG above.

We want to estimate the **direct** causal effect of X on Y . What is the minimum adjustment set? (2p)

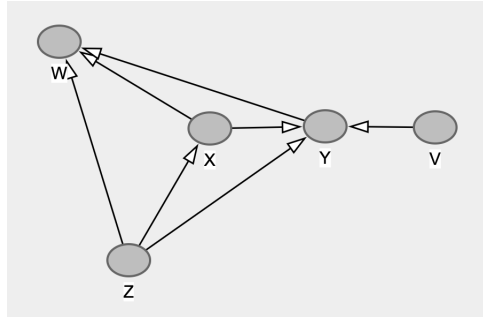
Design a **complete** model, in math notation, where the outcome Y is a count $0, \dots, \infty$ (however, the mean and the variance differs significantly!) **Include the variable(s)** needed to answer the above question. Also add, what you believe to be, **suitable priors on all parameters**. State any assumptions concerning your likelihood and the variables you use! (5p)

**Question 7 :**

(7p) See the DAG above.

We want to estimate the **direct** and **total effect** of X on Y . What is the minimum adjustment set for the **direct** and the **total** effect, respectively? (2p)

Design a **complete** model for estimating **the direct effect**, in math notation, where the outcome y is a real number (i.e., $\mathbb{R}$). **Include the variable(s)** needed to answer the above question. Also add, what you believe to be, **suitable priors on all parameters**. State any assumptions concerning your likelihood and the variables you use! (5p)

**Question 8 :**

(7p) See the DAG above.

We want to estimate the **direct effect** of X on Y . What is the minimum adjustment set? (2p)

Design a **complete** model for estimating **the direct effect**, in math notation, where the outcome y is a real number (i.e., $\mathbb{R}$). **Include the variable(s)** needed to answer the above question. Also add, what you believe to be, **suitable priors on all parameters**. State any assumptions concerning your likelihood and the variables you use! (5p)

Question 9 :

(8p) List and explain the four main benefits of using multilevel models.